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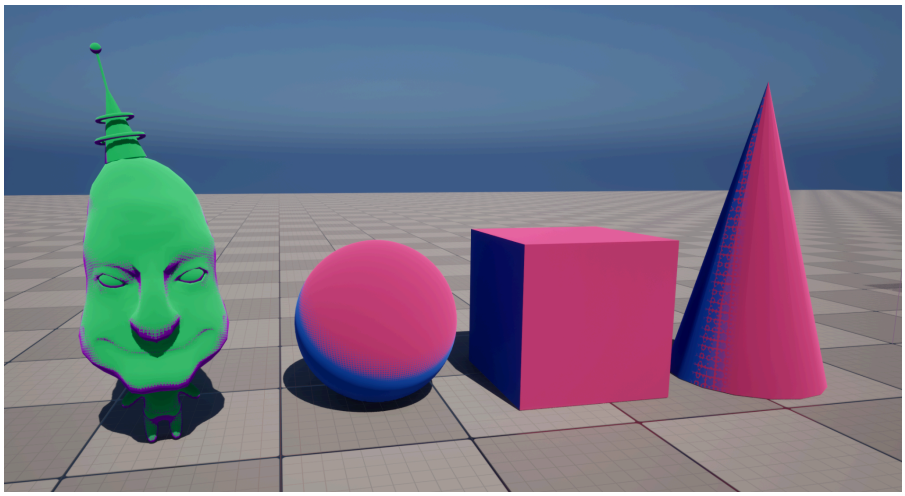
IMGD4099: Real-Time VFX for Games and Interactive Media Debrief

Starting this course, I knew I wanted to work on learning shaders since I hadn't done much work with them at all before. The majority of the learning I did this term was through tutorials, occasionally modifying what came from the tutorial.

LED Screen Material

This LED screen material was the first tutorial that I followed this term. I used Nikolay Shamaev's tutorial on YouTube: https://youtu.be/4cJXZpAbxcg?si=0YQWHDKb4qcGnR4-. The tutorial was straight-forward and easy to follow. I didn't make any modifications to this material since it was my first go at doing work in the material graph editor beyond just connecting textures.

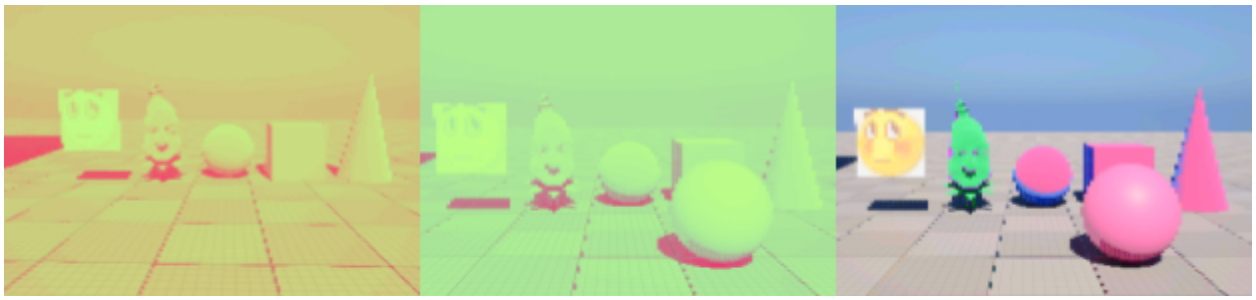
Stylized Comic Material



This stylized comic material was the second thing that I started work on this term. I used Unreal Engine's tutorial on YouTube: https://youtu.be/rpmm3PPxJzM?si=7uvd0tIGPT74CVt0. The main video is less-so a true tutorial and more video briefly viewing the material graph nodes. The description included a link to the same video with extra tutorials and documentation to the Unreal Engine community website. This tutorial from the community was also helpful for the video: https://dev.epicgames.com/community/learning/tutorials/J3ea/unreal-engine-deconstructing-a-stylized-comic-shader. Overall the video is not easily understandable for people new to shaders. It took a bit of time to work through the video even though it's around 2 minutes.

The tutorial originally was going for a crosshatching effect, but I had difficulty with getting the hatches to properly line-up/tile. Instead, I decided to use dots to create a half-tone effect. The tutorial also keeps the material to black and white. To add color, I added a Lerp node between two colors, and the alpha being the output of the tutorial.

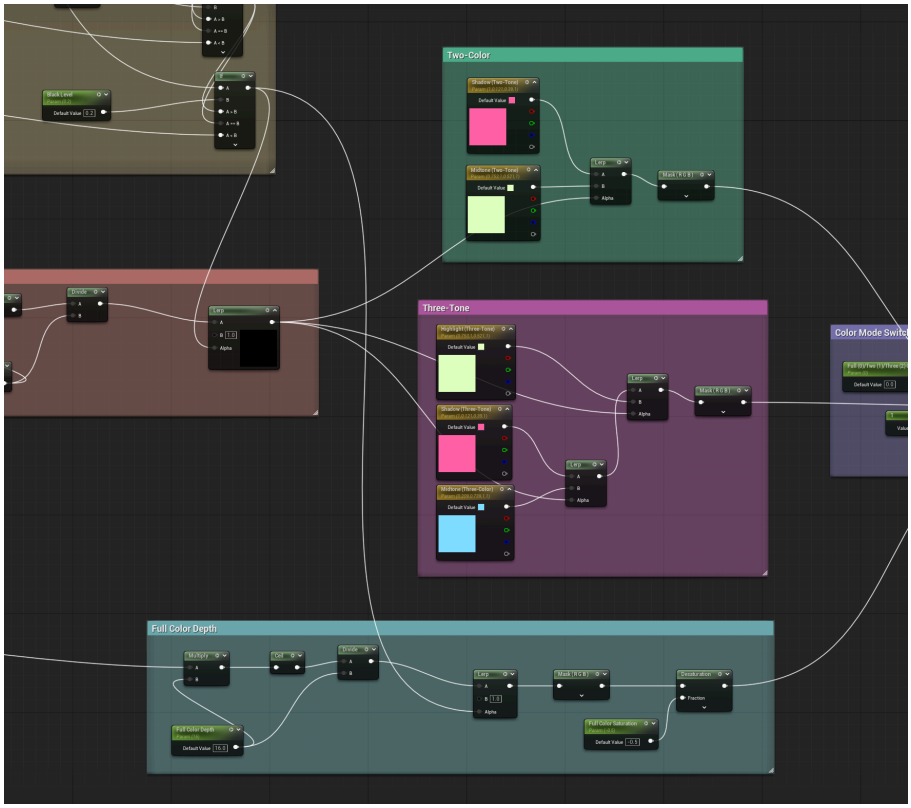
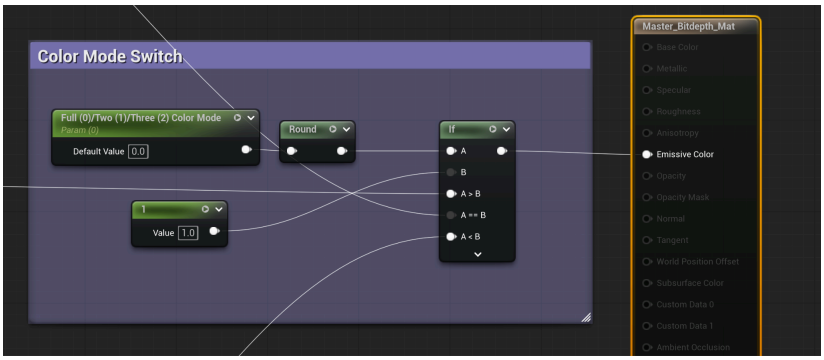
Bitmap Post-Process Material



The bitmap post-process material is my favorite thing that I've worked on this term. I used Nils Gallist's tutorial for creating a realtime bitmap shader on YouTube: https://youtu.be/pwr-H1hro4s?si=y5jNKeHPFQ8tAa0z.

The tutorial shows how to create the effect with two colors, but I wanted to add more options. The two tone color nodes in the tutorial use a Lerp between two vector parameter nodes and use color depth for the alpha. To allow for three colors, I copied the two tone nodes and added a third vector parameter to Lerp between the output of the first two colors and the new third color. The alpha uses the same color depth.

For the full-color mode, I used a modified version of the color depth nodes going into the two/three color modes. The output of the color depth node got plugged into the A channel of a Lerp node, and the output of the bitdepth nodes were plugged into the alpha channel. This added the full-color effect with the bitmap effect. On its own, the colors were more desaturated than I liked, so I used a desaturation node with a scalar parameter to resaturate the colors.



To allow for switching between the different color modes, I used a scalar parameter and an if-statement. To make the mode switching more clear with a scalar, I used a round node to round the scalar output to an integer. I initially had difficulty connecting the color modes into the if-statement. To fix this, I used the component mask node to convert the outputs of the two-tone and three-tone outputs to match the data coming from the full color output.